Phase 3: Performance and Power Simulation and Optimization of LPN-1

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Computer Architecture and Design (MSCS-531-M40)

11/7/2026

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This report looks at the performance and energy of LPN-1 (the low power sensor node core from Phases 1 and 2) and evaluates three optimizations. All the numbers are from runs with gem5 v25.1.0.1. The energy values are modelled estimates derived from the run statistics and assumed coefficients so the ratios between the different configurations are more robust than the absolute values.

Simulation Experiments and Metrics

I measured five metrics: IPC (instructions committed per cycle), execution time (cycles / clock frequency), EPI (total energy / instructions), energy per frame (the energy to process one 256 sample frame including the idle time until the next frame), and EDP (energy x execution time, which penalizes the slow, low energy points). Total energy is the sum of the core, L1, memory and leakage values from energy.py.

I did three experiments with the four kernels from Phase 2 (FIR, feature extraction, CRC-16 and the full pipeline) which made 24 gem5 runs in total. Experiment 1 used a baseline of 0.8V at 50MHz with 8KiB L1 caches. Experiment 2 tested three DVFS points (1.0V/100MHz, 0.8V/50MHz and 0.6V/20MHz). Experiment 3 varied each L1 cache from 1 to 8KiB with cache access energy scaling with the square root of the size and cache leakage scaling linearly with size. For the optimization study we assumed the node processes one pipeline frames every 100ms and then sleeps.

Simulation Results

Table 1
Baseline Results at 0.8 V and 50 MHz (8 KiB L1 Caches)

Kernel	Instructions	Cycles	IPC	Time (ms)	Energy (µJ)	EPI (pJ)	Power (µW)
FIR	721,254	1,188,876	0.607	23.8	15.9	22.0	667
Feature	163,025	227,904	0.715	4.6	3.1	19.0	679
CRC-16	706,261	1,334,643	0.529	26.7	16.4	23.2	614
Pipeline	1,417,427	2,381,086	0.595	47.6	31.1	21.9	653

Note. For the pipeline kernel, energy splits into core 68%, L1 caches 22%, and leakage 10%; memory energy is under 0.01 µJ.

IPC is in the range of 0.53 to 0.72 (well below the single-issue limit of 1), as you'd expect for a very short in-order pipeline. L1 miss rates are at most 0.06%, so memory stalls aren't the cause of the gap. I haven't separated out the various causes of pipeline stalls. Energy per instruction is 19-23pJ and the total is core-dominated.

Table 2
DVFS Results for the Pipeline Kernel

Operating point	Time (ms)	Energy (µJ)	EPI (pJ)	Power (µW)	EDP (µJ·ms)
1.0 V / 100 MHz	23.8	45.5	32.1	1,910	1,083
0.8 V / 50 MHz	47.6	31.1	21.9	653	1,480
0.6 V / 20 MHz	119.1	22.7	16.0	190	2,700

Note. Cycle counts are identical (2,381,086). The other three kernels show the same ordering.

Cycles are independent of frequency so time is proportional to 1/f. Dynamic energy should scale with the square of the voltage (Hennessy & Patterson, 2019) which would reduce it by 64% at 0.6V. Total energy only reduces by 50% (from 45.5 to 22.7µJ) as the run is five times longer and leakage increases from 1.7 to 6.9µJ. That's consistent with the V^2 trend we'd expect plus the increase in leakage. EDP is best at 100MHz so the optimum point will depend on whether you care more about energy or speed.

Table 3
L1 Cache Size Sweep (0.8 V, 50 MHz)

L1 size	FIR D-misses	FIR time	FIR energy	Pipe D-misses	Pipe time	Pipe energy
1 KiB	3,211	+0.26%	−14.9%	1,776	+0.17%	−15.0%
2 KiB	13	0.00%	−12.3%	26	0.00%	−11.9%
4 KiB	13	0.00%	−7.3%	13	0.00%	−7.0%
8 KiB	13	0.00%	0.0%	13	0.00%	0.0%

Note. Changes are relative to 8 KiB. Feature and CRC kernels change by at most 0.01% in time and save 11.6% and 15.5% energy at 1 KiB. D-misses = data-cache misses.

Smaller caches use far less power with virtually no penalty. Reducing the 8KiB caches to 1KiB saves 15% in power and incurs at most a 0.26%-time penalty (even though FIR data misses increase from 13 to 3,211). Each additional miss costs roughly 1 cycle for FIR and 2.4 cycles for the pipeline, since the 10 ns on-chip memory takes 1 cycle or less. The benefit is there unless a miss-refill costs more than about 0.74nJ (FIR) or 2.6nJ (pipeline), which is around 30 or 100 times the 25pJ that I assumed. This result is predicated on the fast SRAM that we assumed in Phase 1.

Analysis and Optimization

The first bottleneck is idle energy rather than processing. At one frame every 100ms the baseline is spending 6.9 of the 9.2 µJ per frame idling (Table 4) as clock gating leaves all the leakage on. I applied four steps in order and only kept each one if the energy went down.

Step S1: put the core and caches into power gated sleep with SRAM retention between frames (using the 0.575 retention factor from Machetti et al. 2024 and a wake-up cost of 0.5 µJ as I assumed). Step S2: power only one of the four 32KiB banks as the program and data together require less than 2KiB. Step S3: use 1KiB L1 caches (the smallest size that satisfies the 2% slowdown requirement across all kernels). Step S4: pick the DVFS point that gives the lowest frame energy. Running fast and sleeping (100MHz) gives 3.06 µJ per frame; 50MHz gives 2.41

µJ and 20MHz gives 1.94 µJ. Slow and low voltage wins in this case as sleep power is non-zero and the dynamic energy is reduced quadratically with voltage.

Table 4

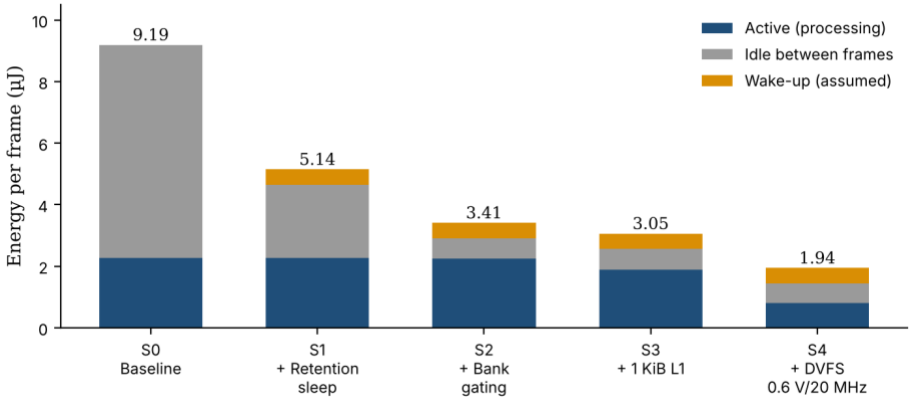
Optimization Steps for the Pipeline Kernel at One Frame per 100 ms

Configuration	Active (µJ)	Idle (µJ)	Wake (µJ)	Total (µJ)	Power (µW)	Frame time / saving
S0 Baseline (1.0 V/100 MHz, clock-gated idle)	2.27	6.92	0.00	9.19	91.9	1.2 ms / 0%
S1 + Retention sleep	2.27	2.37	0.50	5.15	51.4	1.2 ms / 44.0%
S2 + Bank gating (1 of 4)	2.24	0.67	0.50	3.41	34.0	1.2 ms / 63.0%
S3 + 1 KiB L1 caches	1.89	0.67	0.50	3.06	30.5	1.2 ms / 66.8%
S4 + DVFS (0.6 V/20 MHz)	0.80	0.64	0.50	1.94	19.4	6.0 ms / 78.9%

Note. Retention sleep, bank gating, and the wake-up cost are analytic estimates; gem5 does not simulate them.

Figure 1

Energy per Frame After Each Optimization Step



Note. Totals are printed above the bars. Values are from Table 4.

Final Performance and Power Metrics

The optimized node uses 1.94µJ per frame (down from 9.19µJ) which is a 78.9% saving. Average power drops from 91.9 to 19.4µW. Active energy per instruction falls from 32.1 to 11.3pJ. Frame time increases from 1.2 to 6.0ms (6% of the 100ms period) so the deadline is still met. With a nominal 225mAh, 3V coin cell (approx. 2430J) and the processor alone this takes battery life from around 0.8 to 4.0 years.

The savings are a function of the duty cycle: 53.9% at 1 frame per 10ms, 78.9% at 100ms and 88.9% at 1s. The wake-up cost is 26% of the final energy so that must be measured in a real design. The compressed build is still a mystery: its fetch accesses went up in gem5, but the pipeline kernel time only changed by -0.1% so I've left it in for the 23% smaller code size. All of this of course depends on the assumed coefficients; the next step is to calibrate those against silicon data.

References

Hennessy, J. L., & Patterson, D. A. (2019). *Computer architecture: A quantitative approach* (6th ed.). Morgan Kaufmann.

Machetti, S., Schiavone, P. D., Müller, T. C., Peón-Quirós, M., & Atienza, D. (2024). *X-HEEP: An open-source, configurable and extendible RISC-V microcontroller for the exploration of ultra-low-power edge accelerators* (arXiv:2401.05548). arXiv. https://arxiv.org/abs/2401.05548